Library Management System

User Documentation

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Course: Technology and Methodology of System Development (BAI0168)
Website: https://thex006.github.io/library-book-manager/

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1. Purpose of the application

The Library Management System is a simple website that helps a small library keep its records in one place. A lot of small libraries, such as school or club libraries, still write everything on paper or in a spreadsheet. That makes it hard to know which books the library has, who the readers are, and who has borrowed what.

This application lets the librarian store and manage:

- the books of the library,
- the authors who wrote them,
- the categories the books belong to (for example Fiction or Fantasy),
- the readers, who are the members of the library,
- the loans, which record which reader borrowed which book and when.

You don’t need to install anything. The application runs in a normal web browser, and everything you enter is saved automatically.

2. How to use the system

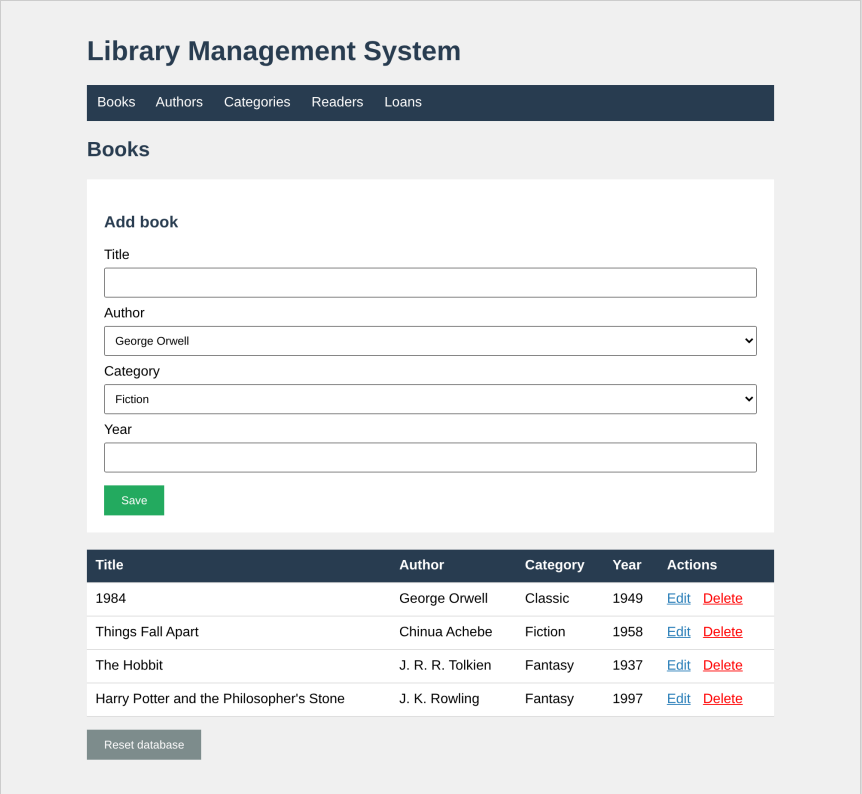
To start, open this address in any web browser (Chrome, Edge or Firefox):

https://thex006.github.io/library-book-manager/

The first page you’ll see is the Books page. At the top there’s a dark menu bar with five links: Books, Authors, Categories, Readers and Loans. Click any of them to go to that page.

All five pages look and work the same way:

- At the top there’s a form, where you add new information or edit existing information.
- Below the form there’s a table that lists everything already saved.
- Next to each row in the table there are two links: Edit (blue) and Delete (red).
- At the bottom there’s a grey Reset database button.



The Books page: the menu at the top, the form, and the list of books

When you open the site for the first time, it already has some example data, so you can try things out straight away.

Your data is saved in your own browser. If you close the page and come back later on the same computer and browser, everything you entered will still be there.

3. Main functions

Page	What you can do there
Books	Add a book, see all books with their author and category, change a book's details, delete a book
Authors	Add, list, edit and delete authors (name and country)
Categories	Add, list, edit and delete categories (for example Fiction, Fantasy, Classic)
Readers	Add, list, edit and delete library members (name, email and phone)
Loans	Record that a reader borrowed a book, see all loans, change the dates, delete a loan when the book is returned

There’s also the Reset database button. It deletes all your changes and puts back the original example data, which is useful if you want to start again from the beginning.

4. Steps for using the application

Here’s a typical way to use the system, from start to finish. It follows the order the information depends on: a book needs an author and a category, and a loan needs a book and a reader.

Step 1: Add an author

1. Click Authors in the menu.
2. Type the author’s name, for example “Toni Morrison”, and their country, for example “USA”.
3. Click Save. The author now appears in the table.

Step 2: Add a category (if you need a new one)

1. Click Categories.
2. Type a name, for example “Poetry”.
3. Click Save.

Step 3: Add a book

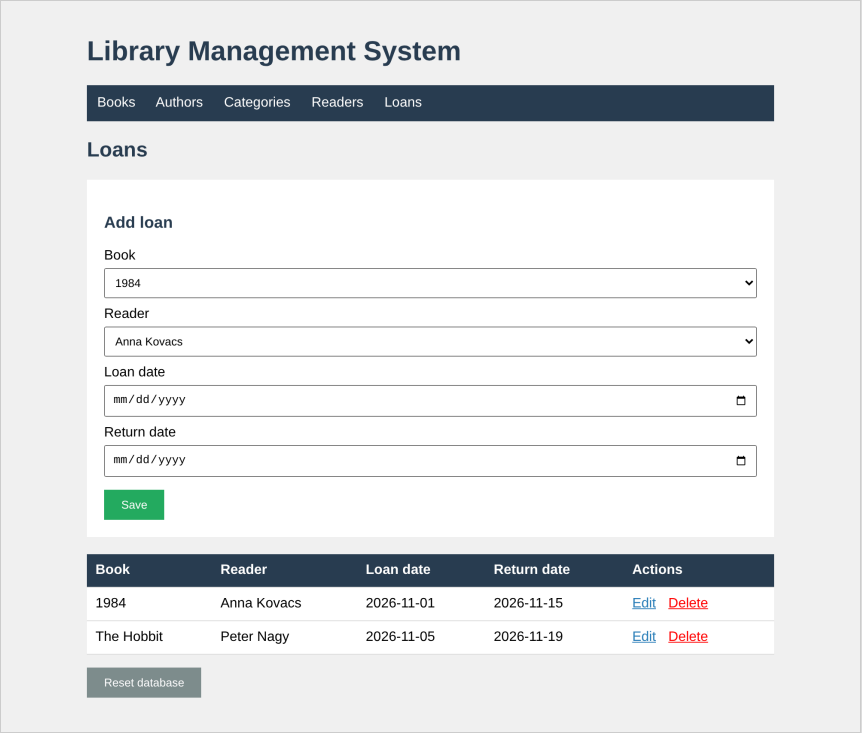
1. Click Books.
2. Type the title, for example “Beloved”.
3. Choose the author from the Author dropdown. The author you added in step 1 will be there.
4. Choose the category from the Category dropdown.
5. Type the year, for example 1987.
6. Click Save. The book appears in the list together with its author’s name and its category.

Step 4: Add a reader

1. Click Readers.
2. Type the reader’s name, email and phone number.
3. Click Save.

Step 5: Record a loan

1. Click Loans.
2. Choose the book from the Book dropdown.
3. Choose the reader from the Reader dropdown.
4. Pick the loan date, and the date the book should be returned.
5. Click Save. The loan appears in the list, so you can see who has which book.



The Loans page: choose a book and a reader, then set the dates

Step 6: When the book comes back

Go to Loans and click Delete next to that loan. It’s removed from the list.

A few tips:

- Fields like Title and Name are required, so the form won’t save if they’re empty.
- If you delete an author or category that is still used by a book, the book stays, but its author or category is shown as “-”.

5. Using the CRUD operations

CRUD stands for Create, Read, Update and Delete. These are the four basic things you can do with the data, and every page supports all four in the same way.

Create (add something new)

Fill in the form at the top of the page and click Save. The new record is saved in the database and appears in the table straight away.

Read (see what is saved)

The table under the form always shows everything that is saved on that page. On the Books page you can see each book’s author and category. On the Loans page you can see the book’s title and the reader’s name.

Update (change something)

1. Click Edit next to the row you want to change.
2. The form title changes to “Edit …”, and the form is filled in with that record’s current data.
3. Change what you need, then click Save.

The table updates with the new information.



Editing a book: the form is filled in with the book’s current data

Delete (remove something)

Click the red Delete link next to the row. The record is removed from the database, and it disappears from the table.

Be careful: deleting can’t be undone. The only way back is the Reset database button, which also removes everything else you have added.

Common problems

- **The page looks plain, with no colours.** Your browser is using an old saved version of the page. Press Ctrl + Shift + R to reload it.
- **My changes disappeared.** The data is saved in the browser you used. If you open the site in a different browser or on another computer, or if you clear your browser data, you’ll see the example data again.
- **The author or category I want isn’t in the dropdown.** Add it first on the Authors or Categories page, then come back to Books.